

TAO (TONY) JIN

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EDUCATION

Cornell University

Master of Engineering in Systems Engineering

Ithaca, NY
Sep. 2025 – May 2026

University of British Columbia

Bachelor of Science: Major in Mathematics

Vancouver, BC
Sep. 2021 – June 2025

- Relevant Coursework: Statistical Learning, Time Series Analysis, Statistical Inference & Hypothesis Testing, Stochastic Calculus, Linear Programming, Algorithms, Relational Databases.

PROJECT EXPERIENCE

Diabetes 30-Day Readmission Risk Modeling | Python, scikit-learn, XGBoost

Sep.–Dec. 2025

- Engineered a 160-feature clinical dataset from 101,766 encounters across 130 U.S. hospitals with patient-level train-calibration-test splits preventing data leakage in a reproducible pipeline.
- Benchmarked logistic regression, random forest, gradient boosting, and XGBoost with cross-validation and hyperparameter tuning for imbalanced classification (AUROC 0.668, AUPRC 0.230); cut Expected Calibration Error 0.342 to 0.004 by isotonic recalibration, added conformal prediction at 90.1% coverage, ran SHAP explainability and subgroup fairness/bias auditing.

LLM Metadata Extraction Pipeline (Kaggle) | Python, NLP, LLM APIs

Jan.–Mar. 2026

- Built a natural language processing (NLP) pipeline extracting structured SDRF metadata across a 71-category taxonomy from 107 publications and 1,659 records; root-caused a 0.575 validation-test gap and benchmarked regex versus LLM structured output via the PRIDE REST API.

Marketplace Retention & Fulfillment Analytics (Olist) | SQL (DuckDB), Window Functions

- Built reusable order-, customer-, and item-level SQL data marts from 99,441 orders, aggregating to order grain before joining to prevent double-counted revenue and protect KPI accuracy; ran cohort analysis, customer segmentation, and retention/churn analysis surfacing a 3.0% repeat-purchase rate, a BRL 7.98M one-time-buyer segment, and an 8.1% late-delivery rate tied to weaker reviews, delivered as an executive scorecard and dashboard.

PROFESSIONAL EXPERIENCE

Aifuku

Data Science Intern

Toronto, ON
Jan. 2025 – May 2025

- Built and documented a Mage ETL pipeline pulling statement PDFs from Google Drive, parsed transaction fields with a Gemini Large Language Model (LLM) API and prompt engineering, and output structured JSON for ledger reconciliation, replacing manual data entry with reproducible automation.
- Wrote and optimized PostgreSQL queries (joins, CTEs, window functions) on a Postgres 15 source-of-truth database of 9 tables, validating order, receipt, inventory, and ledger data feeding Metabase business intelligence (BI) dashboards and KPI reporting; ran exploratory data analysis (EDA) and root cause analysis across 5 linked entities, presented findings and recommendations cross-functionally to non-technical stakeholders, lifting data quality and reporting accuracy.

Grow Investment Group

Institutional Sales Department Intern

Shanghai, China
May 2024 – Aug. 2024

- Engineered Python (pandas) automation scripts cross-referencing multi-source databases for institutional client due diligence and due diligence questionnaire (DDQ) completion, cutting manual data processing time 15% through systematic data validation.
- Authored convertible bond downturn analysis, quantitative market research on macroeconomic, rate, and sentiment data, and product training materials, translating findings into data storytelling for non-technical audiences; drove 9% fewer redemptions, 4% more inquiries, 12% higher conversion.

Sinolink Securities Co., Ltd.

Investment Banking Intern

Shanghai, China
June 2023 – Aug. 2023

- Calculated portfolio risk KPIs (Sharpe, Sortino, Beta, volatility) and correlation analysis for stakeholder risk management reports; ran pre-IPO due diligence (~RMB 460M) with rigorous attention to detail and redesigned reporting workflows, improving team efficiency 8%.

TECHNICAL SKILLS

Programming: Python (pandas, NumPy, scikit-learn, XGBoost, SHAP, Matplotlib), R, SQL (PostgreSQL, Oracle SQL, DuckDB), C++, MATLAB, Git version control, Jupyter

Machine Learning & Statistics: Supervised learning, predictive modeling, regression, feature engineering, model evaluation (precision/recall), model calibration, explainability, fairness/bias auditing, statistical modeling, hypothesis testing, statistical significance, A/B testing/experimental design, causal inference, time series forecasting (SARIMA, Holt-Winters), optimization

Data Engineering & AI: ETL/ELT pipelines (Mage), Apache Spark, Snowflake, data warehousing, data modeling, normalization, data quality, data cleaning, Natural Language Processing (NLP), Large Language Models (LLMs), LLM APIs, prompt engineering, REST APIs

Analytics & BI: Tableau, Power BI, Excel (pivot tables), dashboard development, data visualization, KPI reporting, cohort analysis, segmentation, retention/churn analysis

Business & Communication: Stakeholder communication, cross-functional collaboration, requirements gathering, business acumen, presenting to technical/non-technical audiences, technical writing/documentation, data-driven decisions, working through ambiguity. Languages: Chinese (Native), English (Professional), Japanese (Basic)